Robotics and Autonomous Systems 107 (2018) 100–115

Contents lists available at ScienceDirect

Robotics and Autonomous Systems

journal homepage: www.elsevier.com/locate/robot

Torque control of an active elastic transfemoral prosthesis via quasi-static modelling

Sophie Heins [a,*], Louis Flynn [b], Joost Geeroms [b], Dirk Lefeber [b], Renaud Ronsse [a]

[a] Center for Research in Mechatronics; Institute of Mechanics, Materials, and Civil Engineering; Institute of Neuroscience; and Louvain Bionics; Université catholique de Louvain, Belgium

[b] Robotics and MultiBody Mechanics Research Group, Vrije Universiteit Brussel, Belgium

HIGHLIGHTS

- We developed a bio-inspired, torque-based, controller for the CYBERLEGs prosthesis.
- This control is achieved by means of static modelling and requires no torque sensing.
- A pilot experiment with a transfemoral amputee validated the low-level controller.
- The high-level control combined reference trajectories and bio-inspired impedances.

ARTICLE INFO

Article history:
Received 18 January 2018
Received in revised form 7 May 2018
Accepted 30 May 2018
Available online 15 June 2018

Keywords:
Rehabilitation robotics
Force control
Static modelling
Biologically inspired

ABSTRACT

Developing mechanical devices to restore natural locomotion for transfemoral amputees still raises many challenges. One of them is the development of an efficient control strategy for the prosthesis active joints, with the objective of making it flexible and intuitive to use. This paper focuses on the transfemoral CYBERLEGs Beta-Prosthesis, an actuated elastic ankle–knee prosthesis embedding bio-inspired mechanisms targeting the minimization of its total power consumption. We provide the development of a bio-inspired, torque-based, controller for this device, requiring no torque sensing. Torque control is achieved by means of static modelling of the prosthesis geometrical and elastic relationships. Bench testing of the prosthesis was performed to show that this static model is able to accurately predict the actual behaviour of the device. The model was then used to build the low-level controller of the prosthesis, converting desired torques into reference positions for the joint actuators. Next, a walking experiment with a transfemoral amputee was conducted to assess the feasibility of a torque-based control approach for the prosthesis, with a simple high-level controller combining reference torque trajectories and bio-inspired joint impedances. Results validated the use of our static model for implementing the low-level controller of such an elastic and redundant prosthesis.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Over the past decades, the incidence of lower-limb amputations has risen worldwide. In 2012, it was estimated that 1.9 million people were living with limb loss in the United States, and this number is expected to double by the year 2050 [1,2]. According to [3], 27% of lower-limb amputations are transfemoral – i.e. above-knee – and are caused by conditions including peripheral vascular disease, diabetes, infection, trauma or cancer. Lower-limb loss is a strongly disabling condition and restoring amputees' mobility is essential to improve their quality of life and let them recover more independence. This is even more critical in case of dysvascular amputation, because physical activity can help slow disease progression [4,5]. However, these patients – in particular those suffering from transfemoral amputation – face tremendous difficulties to use conventional prostheses due to the huge physical and cognitive efforts being required to walk with these devices [6,7].

The current tendency for replacing the amputee missing limb is the development of biomimetic prostheses, both for the ankle and knee joints. These prostheses are designed to reproduce healthy joint mechanical properties, and in particular its quasi-stiffness — i.e. the slope of the torque–angle relationship [8,9]. Typically, healthy joint torque and angle walking patterns are taken from well-known datasets, like e.g. Winter data [10], which are averaged

* Correspondence to: Place du Levant, 2 bte L5.04.02, 1348 Louvain-la-Neuve, Belgium
E-mail address: sophie.heins@uclouvain.be (S. Heins).

https://doi.org/10.1016/j.robot.2018.05.015
0921-8890/© 2018 Elsevier B.V. All rights reserved.

biomechanical data, collected from several healthy walkers. The associated ankle torque–angle relationship is typically characterized by a counter-clockwise loop, i.e. a net positive mechanical work over the walking cycle. Consequently, bionic feet require to be equipped with an actuator to provide a net positive energy during a gait cycle, see e.g. [11–14]. In contrast, the normal knee torque–angle relationship shows a clockwise loop, i.e. a global energy dissipation during level-ground walking. Thus a prosthetic knee does not require to be equipped with a power actuator for this specific task. Most advanced knee modules such as the Össur Rheo Knee [15] and the Otto Bock C-leg [16] combine springs and hydraulic, pneumatic, or magneto-rheological dampers. However, active prosthetic knees like the Össur Power Knee [17], the ANGELAA [18] or the MIT CSEA Knee [19] were developed for other locomotion tasks such as stair climbing or sit-to-stand.

On top of that, there have been efforts to develop integrated prostheses for the knee and ankle replacement, e.g. through control interactions between the two joints [20–23]. A couple of passive research prototypes specifically focused on exploiting the energy dissipated by the knee during level-ground walking in order to power the ankle [24,25]. More recently, the CYBERLEGs Alpha-prosthesis was designed by including an Energy Transfer (ET) mechanism between an active ankle and a passive knee module [26,27]. A second version of this device, i.e. the CYBERLEGs Beta-prosthesis, is equipped with active series-elastic knee and ankle joints, so that positive mechanical energy can be provided to both joints [28].

Reducing the cognitive effort required to use a complex device like a transfemoral prosthesis is at least equally challenging as reducing the associated metabolic cost. To this end, efficient control strategies have to be developed and several methods were recently explored in different devices [29]. Some of them are based on the recording of the kinematics of the sound limb and/or other body parts [30,31]. However, the motion of the sensored body parts can be altered if patients show general weakness. Another family of controllers for lower-limb prostheses rely on finite-state machines decomposing the gait cycle into successive phases associated with distinct control laws, both for transtibial and transfemoral prostheses [12,26,32,33]. This state-dependent control law can be bio-inspired: the impedance-based control approaches developed in [19,34] and [35] were inspired from typical torque–angle and torque–velocity relationships. More generally, bioinspired controllers do not necessarily rely on a finite-state machine, but are based on principles identified in healthy humans, such as reflexes, motor primitives and balance control [12,36,37]. Interestingly, the controller developed in [12] for an ankle prosthesis resulted in a prosthetic work adaptation to perturbations comparable to a healthy ankle. Therefore, bio-inspired approaches might potentially make the device impedance comply with the impedance of a biological leg, because it directly derives from the bio-inspired neuro-muscular dynamics.

Most of the control approaches outlined above rely on some compliance between the prosthesis and its environment. When this compliance is explicit and thus relies on an impedance law, this means that the associated controller framework requires a middle level layer guaranteeing proper torque tracking [29]. Moreover, controlling a prosthesis equipped with series elastic actuators requires converting the reference torques provided by such a controller into reference positions for the joint actuators [12,38,39]. This step is not trivial, and strongly depends on the detailed morphology of the device mechanisms. For an integrated transfemoral prosthesis, morphological complexity is typically impacted by non-linear transmissions, series and parallel elasticities, and some degree of redundancy in the actuation chain. The present contribution addresses the challenge of providing a model to achieve the reference torque-to-position conversion for a representative compliant device, namely the CYBERLEGs Beta-prosthesis. This is presented as a first, yet critical, step for the development of compliant control strategies for such a device in order to restore human locomotion. The proposed method developed for this modelling relies on Hooke's law applied to a geometrical analysis and requires no torque sensing. This method reduces the complexity of the system, assuming quasi-static conditions. Moreover it leads to a model that is very light regarding its computational cost, so that it can be readily implemented in a real-time controller. Last but not least, the framework developed here can be straightforwardly implemented for other prosthetic devices, pending that they are made with similar parts as the one described here, i.e. DC actuators, mechanical levers, parallel and series springs, and clutches.

The paper is structured as follows. Section 2 provides a description of the prosthesis mechanisms and the various parameters governing their geometry. Section 3 reviews the static model of the prosthesis, necessary for the torque-to-position conversion. This static model is then experimentally validated on a test bench in Section 4. Thereafter, Section 5 develops the computation of the reference positions as a function of the desired knee and ankle torques. Finally, the obtained low-level controller is validated through a pilot experiment with an amputee in Section 6 and results are discussed in Section 7. The paper ends with a conclusion.

2. Description of the prosthesis

The CAD drawing and the geometrical model of the CYBERLEGs Beta-prosthesis are provided in Fig. 1. The different modules are described in the following sections. A manuscript describing the design of the prosthesis in more details is currently under preparation.

2.1. Knee module

The prosthetic knee holds two main parts: the weight acceptance (WA) mechanism and the knee actuator (KA).

The WA mechanism is highlighted in green in Fig. 1. It holds a spring of high stiffness called the WA spring, and a motor that can engage and disengage this spring. The length of the WA spring is denoted $L_{\mathrm{WA}}(t)$. Moreover, this spring can only be loaded in compression and, when engaged, provides a force only if the knee angle $\theta_k(t)$ is above a configurable threshold θ_{WA}. Otherwise it is at rest and does not provide force to the device. As the name suggests, the WA mechanism is expected to be active during the WA phase of the gait cycle. During this phase, a high joint stiffness is required to handle impact with the ground and carry body weight. After this phase, the spring must be disengaged to allow knee flexion. The controller should make the decision to lock/unlock the WA mechanism at the right moment: locking should take place before heel strike, when the knee angle $\theta_k(t)$ reaches a value slightly below θ_{WA}, and unlocking should happen during the stance phase, when the knee is extending and $\theta_k(t) \approx \theta_{\mathrm{WA}}$.

In parallel, the knee embeds an actuator highlighted in orange in Fig. 1. It is a series elastic actuator (SEA) with a motor driving the position of a carriage along a linear slider. This position is denoted $Z_{\mathrm{KA}}(t)$. The carriage embeds two opposite sets of springs: the baseline (BL) springs (lower part of the carriage) and the extension (EX) springs (upper part). These springs are precompressed: given that L^0_{EX} and L^0_{BL} are respectively the EX and BL springs rest lengths, the design is such that $D < L^0_{\mathrm{EX}} + L^0_{\mathrm{BL}}$. This system provides compliant actuation during knee flexion and extension.

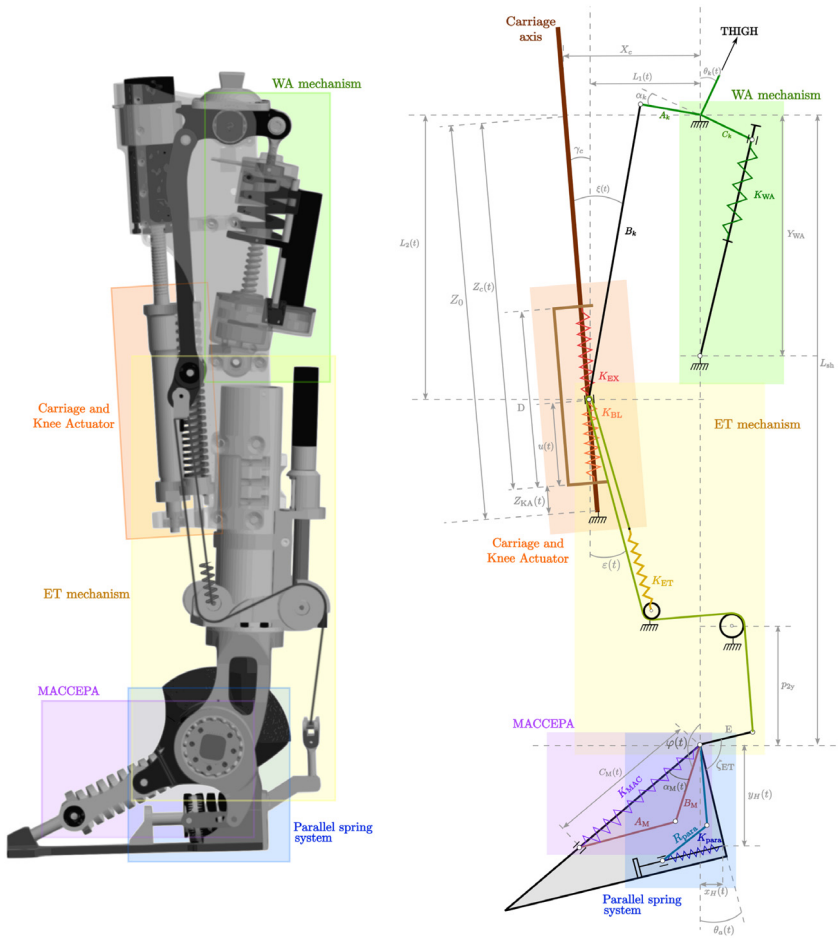


Fig. 1. CAD drawing (left) and Geometrical model (right) of the prosthesis. The knee module includes the WA mechanism (green box) and the knee actuator (orange box). The ankle module holds the parallel-spring system (blue box) and the MACCEPA (purple box). The ET mechanism (yellow box) links the knee joint to the ankle joint. An extended description of the mechanisms is provided in the text, while the list of parameters and their numerical values are provided in Table B.1 in Appendix B. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Ankle module

The prosthetic ankle holds two main parts: a sliding-bar MACCEPA and a parallel spring system.

The MACCEPA is a particular implementation of a variable impedance SEA where it is possible to independently control the actuator compliance and equilibrium position [40,41]. It is highlighted in purple in Fig. 1. It is a SEA containing two rigid bars (A_M and B_M), one sliding-bar embedding a spring (length $C_M(t)$), and three hinge joints in-between. The ankle motor drives the angular position $\varphi(t)$ of the moment arm B_M. By changing this angle, the MACCEPA spring is compressed or decompressed, generating torque at the ankle joint.

The parallel spring system is highlighted in blue in Fig. 1. This passive element allows to reduce the energy consumption of the ankle module by storing energy during dorsiflexion and releasing it during plantarflexion. It contains a spring (length $L_{para}(t)$) that works only in compression. This spring provides a force when the ankle angle $\theta_a(t)$ is above a particular threshold θ_{para}, otherwise it is at rest.

2.3. Ankle–knee energy transfer module

The energy transfer (ET) mechanism is highlighted in yellow in Fig. 1. The aim of this system is to transfer mechanical energy between the knee and the ankle joints. In particular, the energy being dissipated by the knee during level-ground walking could be transferred to the ankle to support the push-off phase. The ET module consists of a cable driven by two pulleys. This cable connects the knee carriage and a lever arm on the back of the ankle. A spring is placed in series in the cable between the first pulley and the knee carriage. During level-ground walking, the ET mechanism should be active during the early swing phase so that the knee flexion extends the cable and pulls the lever arm of the ankle to provide push-off. A motor drives the position of the second pulley (p_{2y}) to lock and unlock the ET mechanism. This system acts as a switch: to lock the ET mechanism, the pulley is moved up with a power screw until the cable is extended. As long as the pulley stays in this extended position, there is a dynamic coupling between the knee and the ankle joints. To unlock the ET mechanism, the second pulley is moved in the opposite direction, i.e. p_{2y} is decreased. In this configuration, the cable is loose and there is no force transmitted across both joints. The controller should make the decision to lock/unlock the ET mechanism at the right moment. For example, during level-ground walking, it should be locked during the stance phase, after the WA phase and as soon as the ankle starts plantarflexing. It should then be unlocked as soon as the ankle starts dorsiflexing, after push-off.

In summary, the knee torque potentially results from the sum of three components: the WA torque τ_{WA} (when this mechanism is engaged); the active torque provided by the knee actuator τ_{KA}, through BL and EX springs; and the torque provided by the ET mechanism τ_{ET}^k (when it is engaged):

$$\tau_k(t) = \tau_{WA}(t) + \tau_{KA}(t) + \tau_{ET}^k(t) \quad (1)$$

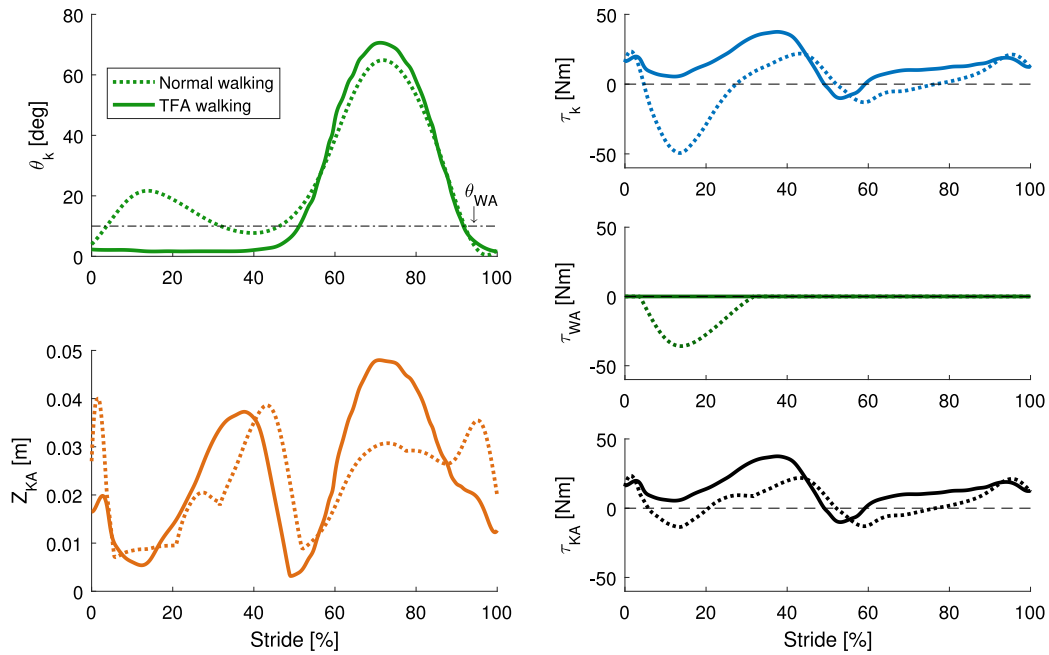


Fig. 2. Static model of the prosthetic knee: knee angle ($\theta_k(t)$), knee torque components ($\tau_k(t)$, $\tau_{WA}(t)$ and $\tau_{KA}(t)$) and corresponding carriage position ($Z_{KA}(t)$), based on Winter data [10] for normal walking and Segal data [42] for transfemoral amputee (TFA) walking (prosthetic limb).

Similarly, the ankle torque is potentially the sum of three independent components, i.e. the parallel spring torque τ_{para}, the active torque provided by the ankle actuator τ_{MAC}, and the torque provided by the ET mechanism τ^a_{ET} (when this mechanism is engaged):

$$\tau_a(t) = \tau_{para}(t) + \tau_{MAC}(t) + \tau^a_{ET}(t) \tag{2}$$

3. Static modelling of the prosthesis

In order to establish the relationships between the total knee and ankle torque and the position of the joint actuators, it is assumed that the system is in static equilibrium. Static modelling of the different modules of the prosthesis is developed in this section.

3.1. Model of the prosthetic knee

3.1.1. Weight acceptance mechanism

When the WA mechanism is engaged, the WA torque depends on the compression force of the WA spring, depending in turn on the knee angle and the WA spring compression limit angle. Given K_{WA} the WA spring stiffness and L^0_{WA} its rest length, this force can be written as:

$$F_{WA}(t) = K_{WA}\big(L^0_{WA} - L_{WA}(t)\big) \tag{3}$$

Moreover, Fig. 1 highlights the geometric relationship between the spring compression $L^0_{WA} - L_{WA}(t)$ and the knee angle $\theta_k(t)$, the exact relationship being given in Appendix A (Eqs. (A.1)–(A.2)). Finally, the WA torque is a knee extension torque, i.e. the product of $F_{WA}(t)$ and its lever arm. It is defined to be negative in our reference frame:

$$\tau_{WA}(t) = -F_{WA}(t)C_k \cos(\theta_k(t) - \rho(t)) \tag{4}$$

where C_k is a distance displayed in Fig. 1 and $\rho(t)$ is developed in Appendix A Eq. (A.3).

Two reference knee angle patterns are provided in Fig. 2: the "Normal walking" pattern is a typical pattern for healthy walking taken from the well-known Winter data [10]; and the "TFA walking" pattern is associated to the prosthetic limb of a transfemoral amputee (TFA) wearing the Mauch SNS prosthesis, reported in [42].[1] According to the activation/deactivation criteria of the WA mechanism given in Section 2.1 and for a compression limit angle $\theta_{WA} = 10.3$ deg, it would be active from 0 to 32% of the gait cycle in case of normal walking. The value of θ_{WA} was fixed to maximize the contribution of the WA mechanism to the knee torque during the WA phase of the gait cycle, while guaranteeing its unlocking before swing phase. The WA torque computed from this knee pattern is displayed in the same figure. It reaches a maximum absolute value of 36 Nm at about 14% of the gait cycle. However, in case of TFA walking, there is no knee flexion during stance phase and consequently, the corresponding WA torque stays equal to zero, since this spring is never under strain.

3.1.2. Energy transfer mechanism

When the ET mechanism is activated, the ET torque at the knee joint $\tau^k_{ET}(t)$ can be calculated as the product between the force in the transfer cable $F_{ET}(t)$ and the corresponding lever arm (see Fig. 1), i.e.:

$$\tau^k_{ET}(t) = F_{ET}(t)\big(L_1(t)\cos\varepsilon(t) + L_2(t)\sin\varepsilon(t)\big) \tag{5}$$

where the relationships between $\varepsilon(t)$, $L_1(t)$, $L_2(t)$, $\xi(t)$ and $\theta_k(t)$ are developed in Appendix A (Eqs. (A.4)–(A.7)).

3.1.3. Series elastic actuator

The KA torque is due to the BL and the EX spring forces ($F_{BL}(t)$ and $F_{EX}(t)$ respectively), both only working in compression. Assuming that the system is in static equilibrium, it can be expressed as:

$$\tau_{KA}(t) = F_{res}(t)A_k \cos\gamma_c\Big(\cos\big(\theta_k(t) - \alpha_k\big) - \sin\big(\theta_k(t) - \alpha_k\big)\tan\big(\xi(t) - \gamma_c\big)\Big) \tag{6}$$

[1] The knee patterns for the sound limb and for the control subject provided in [42] are very similar to Winter data for normal walking. Consequently, it was deemed reasonable to keep Winter data as healthy reference.

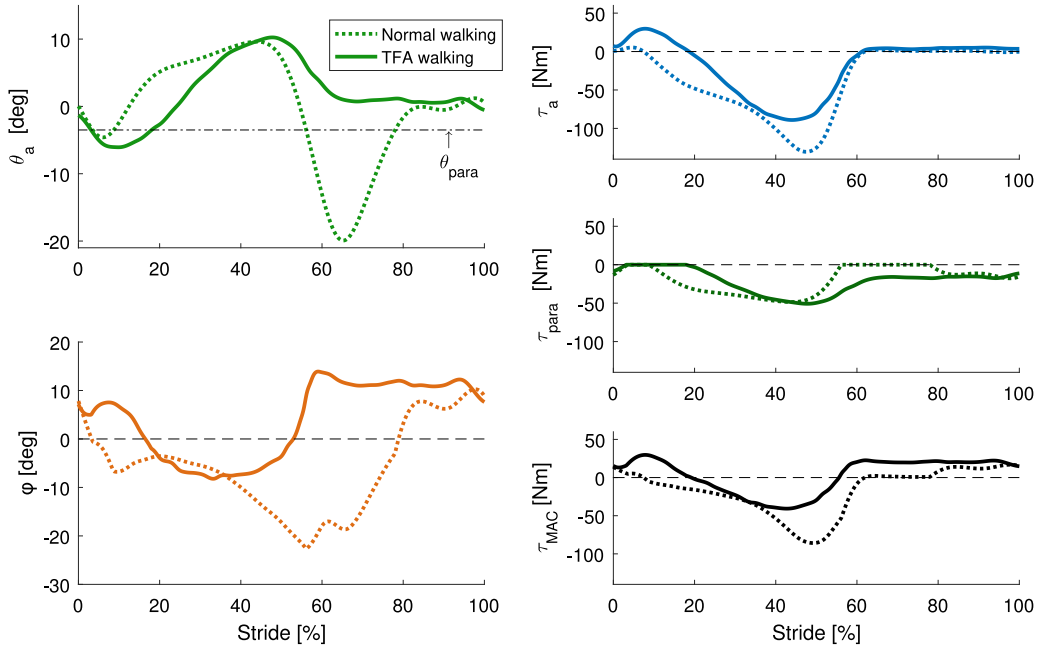


Fig. 3. Static model of the prosthetic ankle: ankle angle ($\theta_a(t)$), ankle torque components ($\tau_a(t)$, $\tau_{para}(t)$ and $\tau_{MAC}(t)$) and corresponding MACCEPA angles ($\alpha_M(t)$ and $\varphi_M(t)$), based on Winter data [10] for normal walking and Segal data [42] for transfemoral amputee (TFA) walking (prosthetic limb).

where $F_{res}(t)$ is the force resulting from the spring forces. This force depends on the length $u(t)$ which in turn depends on the carriage position $Z_{KA}(t)$ (see Fig. 1):

$$u(t) = Z_0 - Z_{KA}(t) - \frac{L_2(t)}{\cos(\gamma_c)} \tag{7}$$

Given the springs precompression, we have:

$$F_{res}(t) = \begin{cases} F_{BL}(t) & \text{if } u(t) < D - L^0_{EX}, \\ -F_{EX}(t) & \text{if } u(t) > L^0_{BL}, \\ F_{BL}(t) - F_{EX}(t) & \text{otherwise.} \end{cases} \tag{8}$$

Given K_{BL} and K_{EX} the BL and EX spring stiffnesses, the spring forces can be expressed as:

$$F_{BL}(t) = K_{BL}\left(L^0_{BL} - u(t)\right) \tag{9}$$

$$F_{EX}(t) = K_{EX}\left(L^0_{EX} - D + u(t)\right) \tag{10}$$

Fig. 2 shows the carriage position profiles $Z_{KA}(t)$ that should be followed to exactly produce the knee torque profiles associated to normal walking [10] and TFA walking [42]. In both cases, the computation accounts for the WA contribution, but disregards the ET one. More details are provided in Section 5.

3.2. Model of the prosthetic ankle

3.2.1. Parallel spring mechanism

The parallel spring torque depends on the force in the parallel spring $F_{para}(t)$, which only works in compression. The compression of the spring depends on the ankle angle $\theta_a(t)$ and is zero when $\theta_a(t)$ is below a particular threshold θ_{para}. According to Fig. 1 and given K_{para} the spring stiffness and L^0_{para} its rest length, we have:

$$F_{para}(t) = K_{para}\left(L^0_{para} - L_{para}(t)\right) \tag{11}$$

where $L_{para}(t)$ is given in Appendix A, as a function of $\theta_a(t)$ (Eqs. (A.8)–(A.12)). Then, the parallel spring torque at the ankle joint can be computed as the product between $F_{para}(t)$ and the corresponding lever arm, i.e.:

$$\tau_{para}(t) = -F_{para}(t)\left(x_H(t)\sin(\nu(t)) - y_H(t)\cos(\nu(t))\right) \tag{12}$$

where $\nu(t)$ is developed in Appendix A Eq. (A.13).

Using again Winter data as reference for normal walking and Segal data for TFA walking, the ankle angle profiles $\theta_a(t)$ over one gait cycle are provided in Fig. 3. The parallel spring torques computed from these angle patterns are displayed in the same figure, for a compression limit angle $\theta_{para} = -3.5$ deg. The value of θ_{para} was fixed to minimize the peak power of the ankle actuator based on normal walking data. The corresponding torque component reaches a maximum absolute value of 48.4 Nm at 44% of the gait cycle for normal walking, and 50.8 Nm at 48% of the gait cycle for TFA walking. Similarly to Fig. 2, the ET mechanism was disregarded in these simulations.

3.2.2. Energy transfer mechanism

The ET torque at the ankle joint $\tau^a_{ET}(t)$ can be computed from the product between the force in the ET cable $F_{ET}(t)$ and its lever arm. If $\theta_{ET}(t) = \theta_a(t) - \zeta_{ET} + \frac{\pi}{2}$ (see Fig. 1), then:

$$\tau^a_{ET}(t) = F_{ET}(t)E\cos(\theta_{ET}(t) + \delta(t)) \tag{13}$$

where $\delta(t)$ is developed in Appendix A Eq. (A.14).

3.2.3. MACCEPA actuator

According to Fig. 1, the MACCEPA torque depends on the angle $\alpha_M(t)$, such that:

$$\tau_{MAC}(t) = C_M(t)F_M(\alpha_M(t), \mathrm{P}) \tag{14}$$

where P is the MACCEPA spring precompression and the length $C_M(t)$ can be written as (by trigonometric relationships):

$$C_M(t) = B_M\cos(\alpha_M(t)) + A_M\sqrt{1 - \left(\frac{B_M}{A_M}\sin(\alpha_M(t))\right)^2} \tag{15}$$

Moreover, $F_M(\alpha_M(t), \mathrm{P})$ is the force perpendicular to $C_M(t)$ and acting at the intersection between this segment and the lever arm A_M. This force depends on the angle $\alpha_M(t)$:

$$F_M(\alpha_M(t), \mathrm{P}) = \frac{K_{MAC}B_M(\mathrm{P} + A_M + B_M - C_M(t))\sin(\alpha_M(t))}{A_M\sqrt{1 - \left(\frac{B_M}{A_M}\sin(\alpha_M(t))\right)^2}} \tag{16}$$

where K_{MAC} is the MACCEPA spring stiffness. The angle of the MACCEPA lever arm with respect to the shank is such that: $\varphi(t) = \theta_a(t) + \alpha_M(t) + \varphi_0$, where φ_0 is the zero position of the controlled angle $\varphi_M(t)$.

Fig. 3 shows the MACCEPA angular position profiles $\varphi(t)$ that should be followed to exactly produce the ankle torque profiles associated to normal walking [10] and TFA walking [42]. In both cases, the computation accounts for the contribution of the parallel spring but disregards the ET one. More details are provided in Section 5.

3.3. Model of the energy transfer mechanism

When the ET mechanism is activated, the knee and the ankle joints are coupled. In this case, there is a force denoted $F_{ET}(t)$ in the ET cable. This force depends on the spring displacement which, in turn, depends on the joint angles. An estimation of $F_{ET}(t)$ in case of normal walking is developed in this section in order to evaluate the potential torque that could result from the ET mechanism at the ankle and knee joints.

As explained in Section 2.3, locking the ET mechanism requires the second pulley to be moved up until the cable is extended and then stays at this fixed position until the ET mechanism is unlocked. Ignoring the portions of the cable that are in contact with the pulleys and considering that the ET spring is at rest at the beginning of the ET phase, the position of the second pulley satisfies the following equation when $t = t_0$, t_0 being the time of the beginning of the ET phase:

$$L_c + L_{ET}(t) = d_1(t) + d_2(t) + d_3 \tag{17}$$

where L_c is the length of the cable, $L_{ET}(t)$ is the spring length, and $d_1(t)$, $d_2(t)$ and d_3 are geometrical parameters developed in Appendix A (Eqs. (A.15)–(A.17)). At time t_0, the ET spring is considered to be at rest: $L_{ET}(t_0) = L^0_{ET}$. Then, during the ET phase, the length of this spring varies with the joint angles according to the same equation, and the corresponding force can be written as:

$$F_{ET}(t) = K_{ET}\left(L_{ET}(t) - L^0_{ET}\right) \tag{18}$$

where K_{ET} is the ET spring stiffness. This spring can only be loaded in extension: if $L_c + L_{ET}(t) > L_c + L^0_{ET}$, the cable is loose and there is no force transmitted through the cable.

Using again Winter data as reference [10] and according to the activation/deactivation criteria of the ET mechanism provided in Section 2.3, the ET phase occurs from 44 to 65% of the gait cycle. Moreover, in order to model the transition to the end of the ET phase, it is assumed that during the 8% stride following the deactivation of the mechanism, the spring stiffness linearly varies from K_{ET} to 0 N/m. The corresponding ET force is provided in Fig. 4. It reaches a maximum of 580 N at 58% of the gait cycle. The corresponding torque at the ankle joint can be derived from Eq. (13), and is shown in the same figure. It reaches a maximum absolute value of 32.2 Nm at 58% of the gait cycle. The corresponding total transferred energy is about 12.2 J. Finally, the power profile of the ankle SEA is the product of the MACCEPA torque computed from Eq. (2) with the time derivative of the MACCEPA angular position $\varphi_M(t)$. Details of the computation of this angle are given in Section 5. The SEA power with ET is compared to the case without ET in Fig. 4. With ET, the positive power peak of the ankle SEA is reduced from 133.8 W to 90.3 W. However, the negative power peak increases from −84.7 W to −188.9 W. This negative impact of the ET mechanism has however to be carefully interpreted. Indeed, our results assume that the Winter torques have to be exactly tracked to produce stable walking, and that they will in turn result in the same kinematics as reported in the Winter data. The assumption that healthy walking is the best way of walking for lower-limb amputees – or even if healthy walking is attainable by amputees – is still questioned, but its falsification will require further investigations, for instance through clinical trials with amputees. These trials are also necessary to better tune the timeframe of the ET locking/unlocking events.

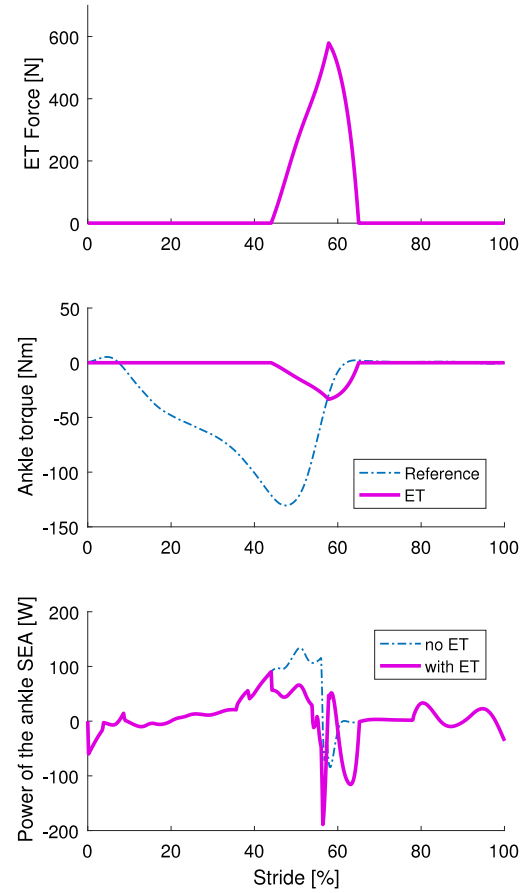


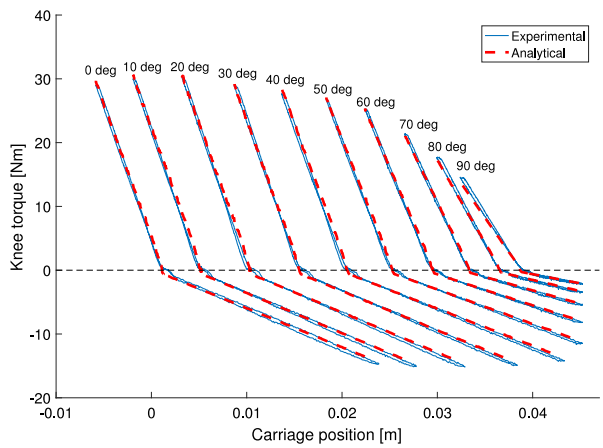
Fig. 4. Estimation of the impact of the ET mechanism, based on Winter data for normal walking [10]: estimated ET force, associated torque at the ankle joint and comparison of the motor power profiles with and without energy transfer at the ankle joint.

4. Validation on a test bench

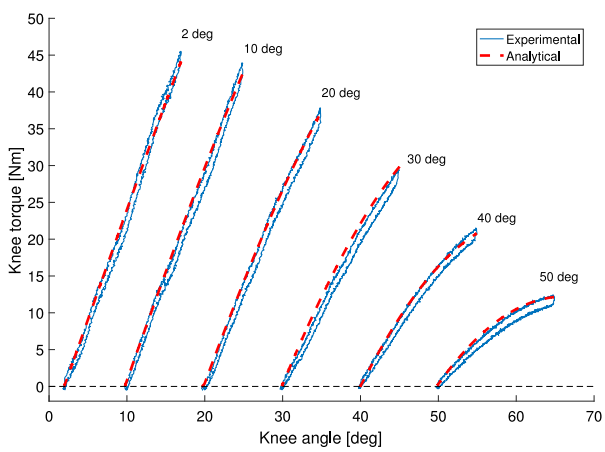
A preliminary validation experiment of the static model of the prosthesis was conducted on a test bench. The knee and ankle modules were tested separately, while no validation of the ET mechanism has been performed so far. For each characterization test, the tested module was fixed on the test bench and the joint output was driven using an external motor while controlling the prosthesis actuator. The motor used to drive the joint angle was a Maxon RE 50 with a GP 62 A gearbox driven by a Maxon EPOS2 70/10 motor controller. It was connected to the joint using a rotary torque transducer (ETH Messtechnik DRBK-200-N). The data acquisition rate was 100 Hz. The measured torque was compared to the analytical prediction.

4.1. Knee actuator

To characterize the knee actuator, the knee angle was fixed to a constant angle and the carriage position was driven in order to generate a torque variation at the knee joint. The ET and WA mechanism were disengaged. Ten trials were performed, each corresponding to a particular value of the knee angle. For each trial,



(a) Knee actuator.



(b) Weight acceptance mechanism.

Fig. 5. Experimental validation of the model of the prosthetic knee. (a) The characterization of the knee actuator was performed for ten different rest positions of the carriage (zero-torque knee angles), from 0 deg to 90 deg. (b) The characterization of the WA mechanism was performed for six different compression limit angles θ_{WA}, from 2 deg to 50 deg, corresponding to six different locked positions of the WA spring Z_{WA} (see Eq. (A.2) from Appendix A).

the carriage position was initially set to the zero-torque position. Each trial included four identical sequences of 20 s each: during the first 10 s, the carriage position $Z_{KA}(t)$ was linearly moved by 7 mm (or until the geometrical limit due to the springs maximum compression was reached) in the negative direction and then moved back to the zero-torque position. During the following 10 s, it was linearly increased by 22 mm (or until the geometrical limit was reached) in the positive direction and then moved back to the zero-torque position. The mean across the four sequences is displayed as result.

The evolution of the knee output as a function of the carriage position is shown in Fig. 5(a), for the ten trials. The predicted torque obtained from Eq. (6) is further reported in the figure. In each trial, there is a good fit between the predicted torque and the experimental one, with a normalized root-mean-square error between 1.6 and 3.4% of the maximum torque. The experimental results show a slight energy loss when the springs are unloaded, so that a small hysteresis is observed in the positive torques (EX springs) as well as in the negative torques (BL springs). The area of these hysteresis loops is between 0.004 and 0.015 J. This hysteresis is likely due to friction effects, and is thus not captured by the present version of the model. These results validate the static model of the knee actuator presented in Section 3.1.3.

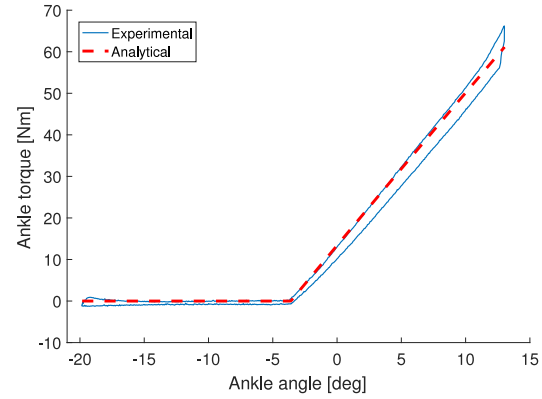


Fig. 6. Experimental validation of the model of the parallel spring of the ankle joint. Measured ankle torque as a function of the driven ankle angle.

4.2. Weight acceptance mechanism

In order to characterize the WA mechanism, the EX and BL springs were removed from the carriage so that the effects of the knee actuator on the knee torque were null. Then, the WA spring was engaged and the knee angle was driven in order to compress the spring and generate a torque variation in the knee joint. Six trials were performed, each of them corresponding to a particular compression limit angle of the WA spring θ_{WA}. At the beginning of each trial, the knee angle was set to θ_{WA}. A trial included four identical sequences of 20 s each: during the first 10 sec, the knee angle was increased linearly by 10 deg, and during the next 10 s it was linearly decreased until θ_{WA}. The mean across the four sequences is displayed as result.

The evolution of the knee torque as a function of the knee angle is shown in Fig. 5(b), for the six trials. The predicted torque from Eq. (4) is further reported in the figure. In each trial, there is a good fit between the predicted torque and the experimental one, with a normalized root-mean-square error between 3.1 and 8.4% of the maximum torque. A hysteresis can again be observed in the experimental results when the WA spring is unloaded, capturing a small energy loss which is not accounted for in the model. The area of these hysteresis loops is between 0.32 and 0.55 J. These results validate the analytical model of the WA mechanism described in Section 3.1.1.

4.3. Ankle module

The actuator of the prosthetic ankle was characterized in [38]. Results showed that the MACCEPA is able to track the desired torque by driving the lever-arm position computed from the static model. Thus, the parallel spring system need to be characterized in order to validate the ankle analytical model described in Section 3.2. During the experiment, the ankle angle was initially set to −20 deg. The trial included seven sequences of 18 s each. In each sequence, the ankle angle was linearly increased until it reached 13 deg, and then decreased back to −20 deg. The mean across the seven sequences is displayed as result.

The evolution of the ankle torque as a function of the ankle angle is shown in Fig. 6. The predicted torque from Eq. (12) is further reported in the figure. Results show a good fit between the experimental torque and the analytical one, although a hysteresis can again be observed in the experimental results during the unloading of the spring, reflecting an energy loss which is not modelled. The normalized root-mean-square error between the experimental signal and the analytical one is 3.3% of the maximum torque and the area of the hysteresis loop is 1.48 J. These results validate the analytical model of the parallel spring system described in Section 3.2.1.

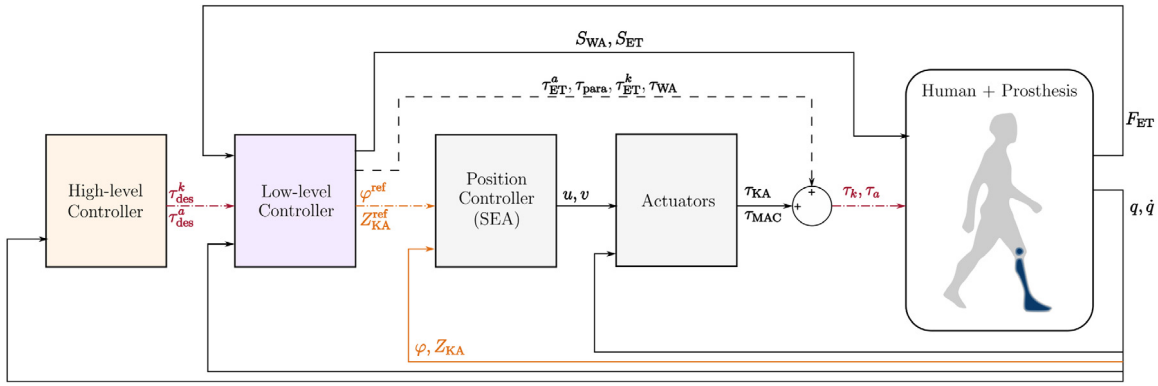


Fig. 7. Block diagram of the prosthesis controller. The low-level controller receives torque references from a high-level controller. It translates these references into position references for the SEA, delivering torques to the amputee and prosthesis complex.

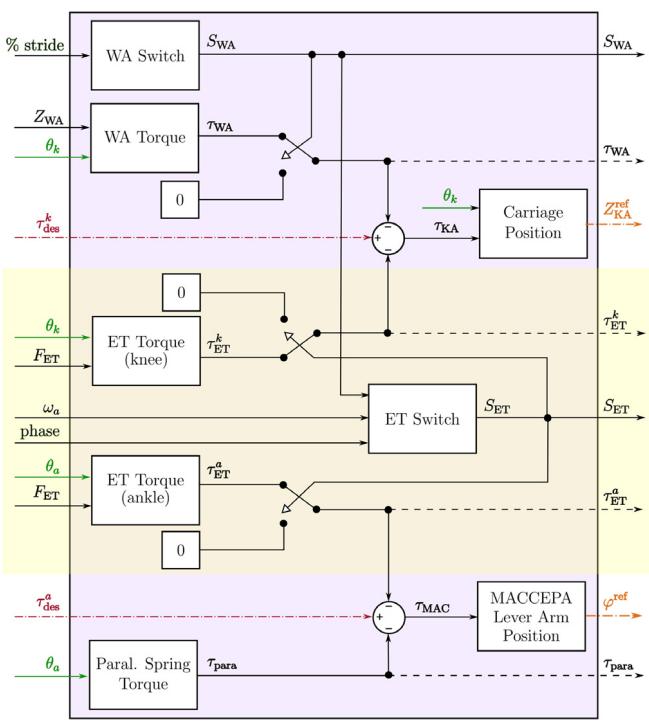


Fig. 8. Block diagram of the prosthesis low-level controller. An extended description of the different blocks is provided in the text. The area highlighted in yellow is specific to the ET mechanism. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5. Low-level controller

Once validated on a test bench, the static model of the prosthesis developed in Section 3 was used to convert reference torques for the knee and ankle joints into reference positions for the prosthesis actuators. This conversion is performed in a so-called low-level controller. The controller architecture of the prosthesis is shown in Fig. 7, the low-level controller being represented by the purple block. The torque references are provided by a high-level controller (orange block). Fig. 8 zooms inside the low-level controller architecture, the region highlighted in yellow being specific to the ET mechanism.

The low-level controller achieves the conversion of the desired joint torques (τ^a_{des} for the ankle and τ^k_{des} for the knee) into reference position for the carriage Z^{ref}_{KA} and the MACCEPA moment arm φ^{ref}. Moreover, it should provide the locking/unlocking command to the WA switch S_{WA} and the ET switch S_{ET}. S_{WA} and S_{ET} are boolean variables. Computing Z^{ref}_{KA} and φ^{ref} requires to estimate the torque contributions from the passive elements. The content of the different blocks of the low-level controller is detailed below.

WA switch.

The WA switch S_{WA} depends on the detected gait percent. It is turned on at heel strike, which happens as soon as the stance phase starts; it is then turned off at the end of the WA phase to allow knee flexion.

ET switch.

The ET switch S_{ET} depends on the detected gait phase (i.e. stance or swing), the WA switch S_{WA} and the measured ankle angular velocity ω_a. It is turned on during the stance phase, as soon as $\omega_a < 0$ while S_{WA} is off; it is then turned off when $\omega_a > 0$.

WA torque.

When S_{WA} is on, the WA torque τ_{WA} at the knee joint is computed from Eq. (4) as a function of the measured knee angle θ_k and the WA spring measured position Z_{WA}; otherwise it is considered to be zero.

ET torques (knee and ankle).

When S_{ET} is on, the ET torques at the knee and ankle joints (τ^k_{ET} and τ^a_{ET}) are computed from Eqs. (5) and (13) respectively, as a function of the measured joint angles θ_k and θ_a and the force F_{ET} measured in the cable; otherwise they are considered to be zero.

Parallel spring torque.

The parallel spring torque τ_{para} is computed from Eq. (12) as a function of the measured ankle angle θ_a.

Once the passive torque components are known, the torques that the SEAs have to provide are equal to the difference between the reference torques provided by the high-level controller and these passive contributions:

$$\tau_{KA}(t) = \tau^k_{des}(t) - \tau_{WA}(t) - \tau^k_{ET}(t) \tag{19}$$

$$\tau_{MAC}(t) = \tau^a_{des}(t) - \tau_{para}(t) - \tau^a_{ET}(t) \tag{20}$$

The desired KA torques corresponding to the knee reference patterns provided in [10] for normal walking and in [42] for TFA

walking are shown in Fig. 2. The desired MACCEPA torques corresponding to the ankle reference patterns are shown in Fig. 3. Remember that, in both figures, the ET mechanism was never engaged.

Carriage position.

The desired force resulting from the EX springs and BL springs forces can be derived from the desired KA torque expressed in Eq. (6):

$$F_{\text{res}}(t) = \frac{\tau_{\text{KA}(t)}}{A_k \cos\gamma_c\big(\cos(\theta_k(t)-\alpha_k) - \sin(\theta_k(t)-\alpha_k)\tan(\xi(t)-\gamma_c)\big)} \tag{21}$$

The length $u(t)$ depends on this force and can be written as:

$$u(t) = \begin{cases} D - \left(\dfrac{F_{\text{res}}(t)}{K_{\text{EX}}} + L^0_{\text{EX}}\right) & \text{if } F_{\text{res}}(t) < -K_{\text{EX}}P_{\text{KA}}, \\ L^0_{\text{BL}} - \dfrac{F_{\text{res}}(t)}{K_{\text{BL}}} & \text{if } F_{\text{res}}(t) > K_{\text{BL}}P_{\text{KA}}, \\ \dfrac{-F_{\text{res}}(t) + K_{\text{BL}}L^0_{\text{BL}} + K_{\text{EX}}(D - L^0_{\text{EX}})}{K_{\text{BL}} + K_{\text{EX}}} & \text{otherwise.} \end{cases} \tag{22}$$

P_{KA} is the spring precompression and equals $L^0_{\text{EX}} + L^0_{\text{BL}} - D$. Finally, according to Fig. 1, the carriage reference position can be calculated as follows:

$$Z^{\text{ref}}_{\text{KA}}(t) = Z_0 - \frac{L_2(t)}{\cos(\gamma_c)} - u(t) \tag{23}$$

where $L_2(t)$ can be computed from Eq. (A.6) of Appendix A, as a function of the measured knee angle $\theta_k(t)$.

The carriage position profiles computed from the knee reference patterns reported in [10] for normal walking and in [42] for TFA walking are shown in Fig. 2. For normal walking, the carriage position varies from 0.71 to 4.00 cm over the gait cycle. For TFA walking, the baseline spring stiffness was increased from 10.7 to 18.0 kN/m in order to be able to exactly produce the desired torque profile. The corresponding carriage position varies from 0.32 to 4.80 cm over the gait cycle.

MACCEPA lever arm position.

The MACCEPA drives the position of the lever arm B_{M}, which depends on the measured ankle angle $\theta_a(t)$ and the MACCEPA angle $\alpha_{\text{M}}(t)$:

$$\varphi^{\text{ref}}(t) = \theta_a(t) + \alpha_{\text{M}}(t) \tag{24}$$

Given the desired MACCEPA torque, $\alpha_{\text{M}}(t)$ can be derived from Eq. (14); however this equation is not easily invertible and would lead to an analytical expression being difficult to solve in a real-time controller. As a consequence, a piecewise polynomial interpolation of the function $\alpha_{\text{M}}(\tau_{\text{MAC}}(t))$ is used to approximate $\alpha_{\text{M}}(t)$, assuming that it would take values between $-45°$ and $45°$ during gait. The piecewise interpolation is decomposed into three third-order polynomials, and the continuous transition between the polynomials is obtained by connecting them through a first order polynomial, in a short interval around their boundary. In this way, the MACCEPA angle is written as:

$$\alpha_{\text{M}}(\tau_{\text{MAC}}(t)) = a_{\text{P}}\big(\tau_{\text{MAC}}(t)\big)^3 + b_{\text{P}}\big(\tau_{\text{MAC}}(t)\big)^2 + c_{\text{P}}\tau_{\text{MAC}}(t) + d_{\text{P}} \tag{25}$$

where the polynomial coefficients a_{P}, b_{P}, c_{P} and d_{P} depends on the desired MACCEPA torque. Their values are given in Appendix C (Fig. C.2). The division in subintervals was optimized by minimizing the root-mean-squared error of the ankle torque over one gait cycle, based on the ankle pattern provided in [10] for normal walking.

The MACCEPA angle profiles $\varphi(t)$ computed from the reference ankle patterns reported in [10] for normal walking and [42] for TFA walking are shown in Fig. 3. In case of normal walking, $\varphi(t)$ varies from -22.6 to 10.3 deg, whereas it varies from -8.2 to 13.9 deg in case of TFA walking.

6. Validation with an amputee subject

The low-level controller developed in Section 5 was finally tested with an amputee subject during level-ground walking. The goal of the experiment was twofold: (i) validate the low-level controller in dynamic conditions and (ii) assess the feasibility of a torque-based control approach for the CYBERLEGs Beta-prosthesis. This second objective is considered as a first step towards the development of any torque-based control strategies for this device, e.g. strategies relying on virtual impedances.

6.1. Method

The prosthesis controller used for the experiment was the controller developed in Section 5 and displayed in Figs. 7 and 8. To achieve the stated objectives, a simple high-level controller (orange block) was implemented to provide the desired joint torques to the low-level controller (purple block). This high-level controller is outlined in the next section.

The experiment was conducted with a transfemoral amputee performing level-ground walking on a treadmill (see Fig. 9). The subject was 59 years old and his amputation was due to osteosarcoma. The prosthesis was used in conjunction with a new socket for the subject. He gave his written, informed consent before starting the experiment under advise of the VUB Medical Ethics Commission (B.U.N. 143201526629) and was free to stop the experiment at any time without providing explanation.

The ET mechanism was considered not safe enough for a walking experiment with an amputee and was thus never activated. Variations in subject kinematics could lead to larger forces than desired, resulting in the ET system not being able to unlock. As a result, the prosthesis ankle would not be able to dorsiflex at the beginning of the swing phase, inducing a high risk of stumbling and resulting in an unsafe situation for the amputee. In the low-level controller shown in Fig. 8, the yellow area can thus be disregarded for this experiment.

The following data were collected during the experiment: the prosthetic knee and ankle angles (θ_k and θ_a), the knee and ankle actuator positions (φ and Z_{KA}), the WA mechanism position (Z_{WA}) and the hip angle (θ_h). The following data were computed by the controller and recorded for off-line analyses: the reference knee and ankle angles (θ^k_{ref} and θ^a_{ref}), the desired knee and ankle actuator positions (φ^{ref} and $Z^{\text{ref}}_{\text{KA}}$), the estimated gait percent and the joint reference torques (τ^k_{ref} and τ^a_{ref}). The data acquisition rate was 100 Hz.

Some of the design parameters were modified in the prosthesis as compared to the version tested in Section 4. The parallel spring compression limit angle θ_{para} and stiffness K_{para} were set to 3.3 deg and 94.2 kN/m respectively. Indeed, the previous values of these parameters brought discomfort to the subject when walking with the prosthesis during preliminary testing. In particular, the subject was not getting sufficient ground clearance. Moreover, the WA spring stiffness K_{WA} was increased to 300 kN/m. In fact, preliminary testing showed that the amputee subject did not trust the WA mechanism and as a consequence did not rely on it. The WA spring was thus replaced with a stiffer one. Moreover, the WA mechanism setpoints were tuned during a short session with the same subject, to guarantee that the locked position was reached at heel strike

Fig. 9. Amputee walking on the treadmill with the CYBERLEGs Beta-prosthesis.

(0% stride) and that the unlocked position was reached at the end of the WA phase (32% stride) to allow knee flexion. As a result, the WA mechanism was switched on at 76% of the stride and switched off at 25% of the stride.

6.2. High-level controller

The prosthesis high-level controller combines reference trajectories and bio-inspired joint impedances, see Fig. 10. For each joint, the desired torque is the sum of a reference torque and an impedance-based torque, i.e.:

$$\tau^a_{\text{des}}(t) = \tau^a_{\text{ref}}(t) + \tau^a_{\text{imp}}(t) \tag{26}$$

$$\tau^k_{\text{des}}(t) = \tau^k_{\text{ref}}(t) + \tau^k_{\text{imp}}(t) \tag{27}$$

where $\tau^a_{\text{ref}}(t)$ and $\tau^k_{\text{ref}}(t)$ are the ankle and knee reference torques and $\tau^a_{\text{imp}}(t)$ and $\tau^k_{\text{imp}}(t)$ are the impedance-based torques. For each joint, the impedance-based torque component only includes a stiffness component and is computed from the difference between the actual joint kinematics and a reference one, i.e.:

$$\tau^a_{\text{imp}}(t) = K_a\left(\theta^a_{\text{ref}}(t) - \theta_a(t)\right) \tag{28}$$

$$\tau^k_{\text{imp}}(t) = K_k\left(\theta^k_{\text{ref}}(t) - \theta_k(t)\right) \tag{29}$$

where K_a and K_k are the ankle and knee virtual stiffness, $\theta^a_{\text{ref}}(t)$ and $\theta^k_{\text{ref}}(t)$ are the joint reference angles and $\theta_a(t)$ and $\theta_k(t)$ are the joint actual angles.

The controller was designed to restore healthy walking. Thus the knee and ankle reference trajectories (i.e. joint torques $\tau^a_{\text{ref}}(t)$ and $\tau^k_{\text{ref}}(t)$ and angles $\theta^a_{\text{ref}}(t)$ and $\theta^k_{\text{ref}}(t)$) were taken from Winter data for slow speed walking [10]. An estimation of the gait percentage was used to synchronize the reference trajectories to the actual gait of the subject. The phase estimation method was based on the Phase Plane Invariant method developed in [43]. It uses the actual hip angle and angular velocity of the prosthetic leg to compute an estimate of the phase angle, which can then be mapped to the gait percent, providing a continuous control signal. The hip angle and angular velocity were measured by two 2D angular rate gyros (ST Microelectronics LPR5150AL). Then, a look-up table was used to map the estimated gait percent to the reference joint torques and angles.

A preliminary walking session was conducted with the amputee for tuning the impedance gains. An iterative approach was used: the subject walked on a treadmill with the prosthesis with given values of K_a and K_k. Then, results were assessed based on the observed kinematic profiles compared to the reference ones and on the participant's comfort. The values of the parameters were modified for the next iteration. The final stiffness values were constant over the gait cycle and equal to $K_k = 0.22$ Nm/rad/kg and $K_a = 2.64$ Nm/rad/kg.

6.3. Results

The amputee was able to walk with the CYBERLEGs Beta-prosthesis and the proposed controller, at a walking cadence of 0.8 stride/sec, corresponding to a forward speed of 0.6 m/s. The recorded data showed very repetitive profiles across the gait cycles. Figs. 11 and 12 show the averaged joint angles over 82 consecutive gait cycles, the corresponding actuator positions and the corresponding joint torques, for the ankle and the knee respectively. The desired joint torques were computed from Eqs. (26) and (27) as a function of the measured joint angles and the recorded reference trajectories, whereas the actual joint torques were computed from inverse kinematics relationships (Eqs. (2), (12) and (14) for the ankle joint and (1), (4) and (6) for the knee joint).

6.3.1. Ankle joint

The observed ankle angle had a profile similar to the reference one, although with less dorsiflexion (see Fig. 11(a)). It reached a maximum of 5.50 deg at 41.4% stride, whereas the reference angle reached a maximum of 8.54 deg at 44% stride. This lead to a parallel spring torque lower than expected: from Eq. (12), the maximum reached angle corresponded to an absolute torque of 7.39 Nm, compared to 17.56 Nm with the reference angle. However, the experimental results showed that the MACCEPA actuator was able to actively compensate for this torque component. Indeed, the ankle motor was able to track the desired position, as shown in Fig. 11(b). The resulting torque at the ankle joint was close to the desired one (Fig. 11(c)). These results show that the prosthesis ankle joint can be properly torque-controlled with the developed low-level controller.

6.3.2. Knee joint

Fig. 12(a) shows that the knee kinematics slightly differed from the reference one. Indeed, the reference profile is characterized by a full joint extension at heel strike followed by stance flexion. However, during the experiment, the knee full extension was only reached after heel stride, i.e. at 28% stride. This behaviour prevented the WA mechanism from reaching its locked position just before heel strike, because the top of the WA spring hit the rigid bar C_k (see Fig. 1). Consequently, the amputee did not lean on the WA mechanism as much as expected, and the WA torque during the WA phase was limited to a maximum absolute value of 13.76 Nm (obtained from Eq. (4)). Results also showed that the knee motor was able to track the desired position (Fig. 12(b)). Although the resulting torque at the knee joint was close to the desired one, a significant difference could be observed between 5% and 18% stride (Fig. 12(c)). This difference was associated with a compression of the EX springs. Due to their high stiffness, small position differences lead to high force differences when an extension torque was desired.

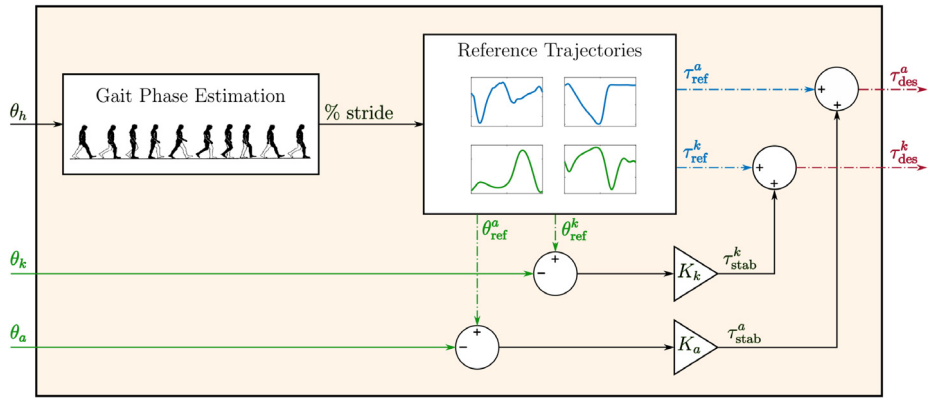


Fig. 10. Block diagram of the prosthesis high-level controller, providing the desired joint torques to the low-level controller. It combines reference position and torque profiles and bio-inspired joint impedances.

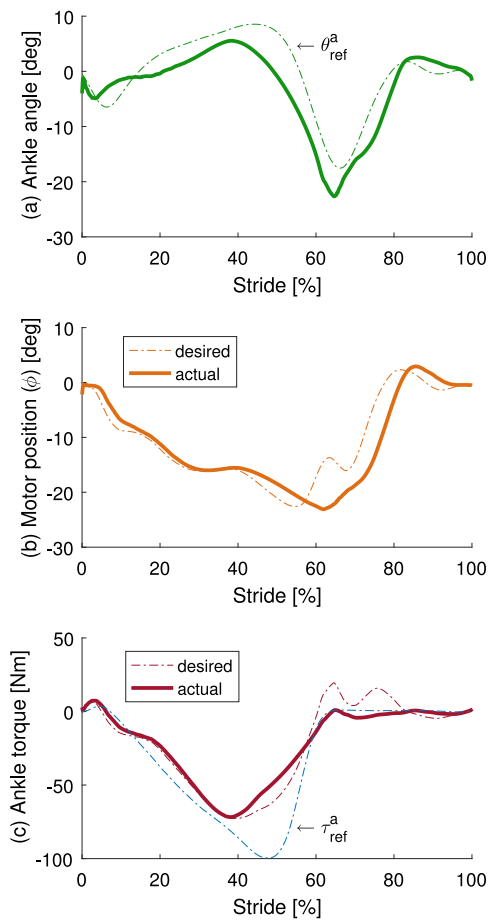


Fig. 11. Experimental validation with the amputee for the ankle joint: (a) averaged ankle angle (θ_a) compared to the reference one (θ^a_{ref}); (b) averaged ankle actuator position (φ) compared to the desired one (φ^{ref}); (c) corresponding ankle torque (τ_a) compared to the desired (τ^a_{des}) and reference ones (τ^a_{ref}). The colour code is consistent with the signals displayed in Figs. 7–9. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

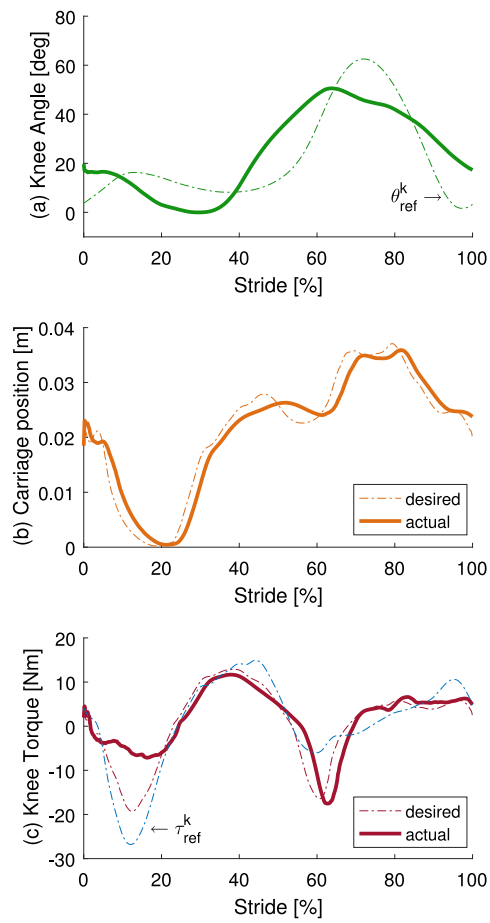


Fig. 12. Experimental validation with the amputee for the knee joint: (a) averaged knee angle (θ_k) compared to the reference one (θ^k_{ref}); (b) averaged knee actuator position (Z_{KA}) compared to the desired one (Z^{ref}_{KA}); (c) corresponding knee torque (τ_k) compared to the desired (τ^k_{des}) and reference ones (τ^k_{ref}). The colour code is consistent with the signals displayed in Figs. 7–9. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

7. Discussion

In this contribution, we derived a static model of the CYBERLEGs Beta-prosthesis, with different torque components acting at the knee and ankle joints. The model was experimentally validated by comparing predicted joint torques with measured ones, obtained during quasi-static experiments on a test-bench. Results showed that the static model accurately predicted the actual behaviour of the prosthesis, although the spring unloadings were associated with slight energy losses (hysteresis) that were not captured in the model. Afterwards, the static model was used to build a low-level controller for the prosthesis. This low-level controller converts

the reference torques for the knee and ankle joints into reference positions for the joint actuators. A walking experiment with a transfemoral amputee was then conducted to assess the feasibility of a torque-based control approach for the prosthesis. A simple high-level controller was implemented to compute the desired joint torques, which combined reference torque trajectories and bio-inspired joint impedances.

7.1. Torque-based control strategy

The walking experiment with the amputee showed that the model derived for the knee and ankle modules is appropriate for implementing a torque-based low-level controller, while there is no torque sensor in the loop. The ankle actuator was able to track the desired position, leading to a torque profile close to the desired one. As for the knee joint, the actuator was also able to track the desired position. Although the resulting torque profile was close to the desired one in flexion, significant differences were observed when an extension torque was desired, due to the high stiffness of the EX springs. However, we believe that these results could be improved with a better use of the WA mechanism that should provide most of the extension torque. According to [42], stance knee flexion is usually not observed on transfemoral amputees' prosthetic leg, because they associate knee flexion with buckling. Extensive training with the prosthesis may improve this behaviour and result in the participant leaning on the WA mechanism. In summary, the experiment showed that the prosthesis knee joint can be efficiently torque-controlled with the developed low-level controller when a flexion torque is desired. However, when an extension torque is desired, the WA mechanism should provide most of the torque in order to accurately track the torque target.

7.2. Impedance-based torque

For the sake of simplicity, we included only a stiffness component in the impedance-based torque of the high-level controller. This compliant torque component allows to deviate from the reference trajectories and compensate for the slight hysteresis that was observed during the test bench experiments. The virtual ankle and knee stiffness K_a and K_k were tuned during a short preliminary session with the amputee. In this section, we compare these values to estimates of the human's knee and ankle equivalent mechanical impedance available in the literature.

The ankle natural mechanical impedance during walking was quantified in [44] using perturbations. According to [8], this method is more reliable and accurate than the concept of quasi-stiffness, i.e. the slope of the torque–angle profile. The ankle stiffness obtained in [44] is approximately constant during swing (0.35 Nm/rad/kg). During stance, it increases linearly from 1 Nm/rad/kg at heel strike to 6.42 Nm/rad/kg at terminal stance. However, the ankle stiffness was not estimated during push-off due to missing data. In our experiment, the ankle virtual stiffness was set to $K_a = 2.64$ Nm/rad/kg, which is thus in the same order of magnitude than the estimated natural stiffness of the ankle joint during walking.

There is currently no quantitative data to estimate the knee natural mechanical impedance during walking using perturbations, although an experimental tool is currently being developed by [45]. A recent study estimated the knee stiffness during walking using kinetic, kinematic and electromyographic measurements [46]. Results showed that the knee stiffness is around 1.39 Nm/rad/kg during swing whereas it reaches a maximum of 6.25 Nm/rad/kg during stance, more precisely during the WA phase. In our experiment, the knee virtual stiffness was set to $K_k = 0.22$ Nm/rad/kg, which is one order of magnitude lower than the estimated natural stiffness of the corresponding human joint during walking.

Nevertheless these results have to be interpreted with care. Indeed, in both joints, most of the desired torque comes from the reference torque ($\tau^a_{\text{ref}}(t)$ and $\tau^k_{\text{ref}}(t)$) and the impedance-based torque component strongly depends on the angle reference trajectory. The tuned stiffness values would probably be different with other angle and torque reference trajectories. Moreover, the tuning session was based on a qualitative assessment, which was deemed sufficient giving the objectives of the experiment. It would however be interesting to compare the amputee's behaviour for different set of impedance parameters in future works. These parameters could take different values in the swing and stance phases, as observed in the natural mechanical impedances [44,46]. A damping component could also be added to the impedance-based torque.

7.3. Adaptation to each subject

The pilot walking experiment was conducted with a single amputee subject, and the prosthetic system was configured for this specific subject. The CYBERLEGs Beta-prosthesis is indeed an adaptive system. The prosthetic design parameters can be adapted to each user's characteristics: the prosthetic shank length can be adjusted to the subject's height and the springs – i.e. their stiffness and rest length – can be changed according to the joint desired quasi-stiffness. The high-level control can also be adapted to the patient's characteristics and targeted behaviour. Indeed, the subject's weight is a multiplicative factor in the computation of the joint desired torque, and the following parameters are configurable: the WA spring locked position Z_{WA}, and thus its compression limit angle θ_{WA}, and the knee and ankle virtual stiffness of the impedance-based torque. Furthermore, the proposed controller is adaptive to the instantaneous walking speed thanks to the real-time estimation of the gait phase, which allows to synchronize the reference trajectories to the actual gait of the subject. Finally, the joint torque and angle reference trajectories can be changed depending on the targeted subject's behaviour. In the pilot walking experiment, Winter data was used with the aim to restore natural locomotion. However, these patterns could be configured to best fit the prosthetic user, e.g. by using reference data for amputee walking such as [42] or by recording the kinematics of the subject's sound limb.

Future work will go deeper into making the prosthesis control adaptive to the subject's behaviour. The reference torque and angle patterns will adapt to the walking speed and to the locomotion tasks being performed (e.g. stair ascending/descending) through the use of primitives combined with adaptive oscillators, similarly as in [37]. Furthermore, the tuning session of the impedance gains was based on a qualitative assessment but it could be automatized, i.e. by implementing a fuzzy tuner using quantitative data, similarly as in [47].

7.4. Energy transfer mechanism

The ET mechanism was considered not safe enough for an experiment with an amputated subject. However, the possible energy that could be transferred from the knee joint to the ankle joint was estimated using Winter's kinematic data. The corresponding power profile of the ankle SEA highlighted the necessity to further study the dynamic couplings between the user, the prosthesis, and the environment, in particular when the ET mechanism is engaged. Indeed, we assumed that the prosthesis joints have to perfectly track the Winter torque, which might be too conservative, and likely unrealistic when both joints are coupled through the ET mechanism. Future experiments are required to (i) study this

dynamic coupling and determine if combining active joints with an ET mechanism can lead to a decrease in both the metabolic effort required to walk with the device and its total electrical consumption; and (ii) optimize the activation/deactivation criteria of the ET mechanism. Comparisons between the predicted and the actual transferred energy will also be done in future works to validate the ET model.

8. Conclusion

This paper provided a first –yet critical –step in the development of a bio-inspired, torque-based controller for a representative compliant transfemoral prosthesis, namely the CYBERLEGs Beta-prosthesis. We derived a static model of this transfemoral prosthesis and validated it through quasi-static experiments on a test-bench. The static model was then used to convert the reference torques for the knee and ankle joints into reference positions for the joint actuators, in a so-called low-level controller. Finally, we assessed the feasibility of a torque-based control approach for the prosthesis during a pilot walking experiment with a transfemoral amputee. A simple high-level controller was implemented to compute the desired joint torques, which combined reference torque trajectories and bio-inspired joint impedances. Results showed that the model derived for the knee and ankle modules is appropriate for implementing the prosthesis low-level controller. This experiment thus paves the way for the development of torque- and/or impedance-based controllers adapting to the user's intention and to the actual task being completed.

This static modelling framework should be of interest for the development of control frameworks for other devices than the one being reported here, mainly if made with the similar parts (levers, springs, and clutches). In our framework, torque references are directly transformed into position references for the device actuators, so that the prosthesis is torque-controlled while having no torque sensor in the device, and thus no need to close the control loop on the torque signal. The experiment with a transfemoral amputee provides an elegant illustration that this simple low-level control approach is appropriate to achieve stable and stationary walking with a simple high-level torque-based control algorithm.

Acknowledgements

This work was supported by the European Union's Seventh Framework Programme (FP7) under grant number FP7-ICT-287894 (the CYBERLEGs collaborative project), the European Community's H2020 Research and Innovation Programme under grant number 731931 (the CYBERLEGs Plus Plus collaborative project), Flanders Innovation and Entrepreneurship (Vlaio, Ph.D. grant awarded to J. Geeroms) and the Belgian F.R.S.-FNRS (crédits aux chercheurs 6809010 awarded to R. Ronsse).

Appendix A. Geometrical relationships of the prosthesis

The parameters used in the geometrical relationships derived in this section are provided in Fig. C.1, which is an extended version of Fig. 1.

A.1. Knee module

A.1.1. Weight acceptance mechanism

The geometrical relationship between the WA spring compression $L^0_{\mathrm{WA}} - L_{\mathrm{WA}}(t)$ and the knee angle $\theta_k(t)$ is given by:

$$L^0_{\mathrm{WA}} - L_{\mathrm{WA}}(t) = Z^0_{\mathrm{WA}} + Z_{\mathrm{WA}}(t) - \sqrt{(C_k \cos\theta_k(t))^2 + (Y_{\mathrm{WA}} - C_k \sin\theta_k(t))^2} \quad (A.1)$$

where Z^0_{WA} is the unlocked position of the WA spring and $Z_{\mathrm{WA}}(t)$ is its actual position. The relationship between the position of the WA spring and the compression limit angle is given by:

$$Z^0_{\mathrm{WA}} + Z_{\mathrm{WA}}(t) = \sqrt{(C_k \cos\theta_{\mathrm{WA}})^2 + (Y_{\mathrm{WA}} - C_k \sin\theta_{\mathrm{WA}})^2} \quad (A.2)$$

The angle $\rho(t)$ can be written as:

$$\rho(t) = \arctan\left(\frac{C_k \cos\theta_k(t)}{Y_{\mathrm{WA}} - C_k \sin\theta_k(t)}\right) \quad (A.3)$$

A.1.2. Energy transfer mechanism

The relationships between $\varepsilon(t)$, $L_1(t)$, $L_2(t)$, $\xi(t)$ and $\theta_k(t)$ are written as:

$$\tan\varepsilon(t) = \frac{X_c - p_{1x} - L_2(t)\tan(\gamma_c)}{L_{sh} - p_{1y} - L_2(t)} \quad (A.4)$$

$$L_1(t) = A_k \cos(\alpha_k - \theta_k(t)) + B_k \sin(\xi(t) - \gamma_c) \quad (A.5)$$

$$L_2(t) = A_k \sin(\alpha_k - \theta_k(t)) + B_k \cos(\xi(t) - \gamma_c) \quad (A.6)$$

$$\sin\xi(t) = \frac{1}{B_k}\left(X_c \sin(\frac{\pi}{2} - \gamma_c) - A_k \sin(\frac{\pi}{2} - \gamma_c + \alpha_k - \theta_k(t))\right) \quad (A.7)$$

A.2. Ankle module

A.2.1. Parallel spring mechanism

The instantaneous length of the parallel spring $L_{\mathrm{para}}(t)$ can be written as:

$$L_{\mathrm{para}}(t) = \sqrt{(x_M(t) - x_H(t))^2 + (y_M(t) - y_H(t))^2} \quad (A.8)$$

When $\theta_a(t) = \theta_{\mathrm{para}}$, the spring first attachment point position with respect to the ankle joint $\left(x_H(t), y_H(t)\right)$ is known and equal to $\left(x^0_H, y^0_H\right)$. Otherwise it depends on the ankle angle $\theta_a(t)$:

$$\begin{cases} x_H(t) &= \sin(\theta_a(t))\sqrt{(x^0_H)^2 + (y^0_H)^2} \\ y_H(t) &= \cos(\theta_a(t))\sqrt{(x^0_H)^2 + (y^0_H)^2} \end{cases} \quad (A.9)$$

At the other side, the spring is attached to a connection point $\left(x_M(t), y_M(t)\right)$, which corresponds to the intersection of:

- the extremity of the rigid bar R_{para}, whose other extremity $\left(x_R, y_R\right)$ is connected to the shank; this bar rotates around $\left(x_R, y_R\right)$.
- the sliding bar which is aligned with the sole of the foot.

Thus, $\left(x_M(t), y_M(t)\right)$ must satisfy the following equations:

$$R^2_{\mathrm{para}} = \left(x_M(t) - x_R\right)^2 + \left(y_M(t) - y_R\right)^2 \quad (A.10)$$

$$y_M(t) = \frac{y_F(t) - y_H(t)}{x_F(t) - x_H(t)} x_M(t) + y_F(t) - \frac{y_F(t) - y_H(t)}{x_F(t) - x_H(t)} x_F(t) \quad (A.11)$$

Also, when $\theta_a(t) = \theta_{\mathrm{para}}$, the position of the extremity of the sliding bar $\left(x_F(t), y_F(t)\right)$ is known and equal to $\left(x^0_F, y^0_F\right)$. Otherwise it depends on the ankle angle $\theta_a(t)$:

$$\begin{cases} x_F(t) &= \sin(\theta_a(t) + \kappa)\sqrt{(x^0_F)^2 + (y^0_F)^2} \\ y_F(t) &= \cos(\theta_a(t) + \kappa)\sqrt{(x^0_F)^2 + (y^0_F)^2} \end{cases} \quad (A.12)$$

where $\kappa = \arctan\left(\frac{x^0_F}{y^0_F}\right) - \theta_{\mathrm{para}}$.

Table B.1
Descriptions and values of the analytical model parameters.

Parameter	Description	Value for static model
K_{WA}	WA spring stiffness	200 kN/m
L^0_{WA}	WA spring rest length	0.038 m
Z^0_{WA}	WA spring unlocked position	0.127 m
Z_{WA}	WA spring locked position	26.2 mm
K_{BL}	BL spring stiffness (2 springs)	10.7 kN/m
L^0_{BL}	BL spring rest length	0.064 m
K_{EX}	EX spring stiffness (2 springs)	89 kN/m
L^0_{EX}	EX spring rest length	0.032 m
C_k	Length of the WA lever arm	0.03 m
A_k	Length of the KA lever arm	0.03 m
B_k	Length of the rigid bar	0.15 m
Y_{WA}	Vertical distance between the knee joint and the WA axis	0.155 m
X_c	Horizontal distance between the knee joint and the carriage axis	0.035 m
γ_c	Orientation of the carriage axis	5 deg
α_k	Angle of the KA lever arm when $\theta_k(t) = 0$ deg	20 deg
D	Length of the carriage	0.095 m
Z_0	Zero-position of the carriage	0.224 m
L_{sh}	Length of the shank	0.4 m
(p_{1x}, p_{1y})	Position of the first pulley's centre	(0.005; 0.095)
(p_{2x}, p_{2y})	Position of the second pulley's centre	(0.025; p_{2y})
R_p	Radius of the pulleys	0.01 m
K_{ET}	Stiffness of the cable	100 kN/m
θ_{para}	Parallel spring compression limit angle	−3.5 deg
K_{para}	Parallel spring stiffness	103 kN/m
L^0_{para}	Parallel spring compression rest length	0.025 m
K_{MAC}	MACCEPA spring stiffness	130 kN/m
(x^0_H, y^0_H)	Position $(x_H(t), y_H(t))$ when $\theta_a(t) = \theta_{para}$	(0.003; 0.046)
(x_R, y_R)	Position of the extremity of R_{para} fixed to the shank	(0.008; 0.045)
(x^0_F, y^0_F)	Position $(x_F(t), y_F(t))$ when $\theta_a(t) = \theta_{para}$	(0.028; 0.047)
R_{para}	Length of the rigid bar of the parallel spring system	0.03 m
A_M	Length of the drive side moment arm of the MACCEPA	0.05 m
B_M	Length of the spring side moment arm of the MACCEPA	0.05 m
P	Precompression of the MACCEPA spring	0.007 m
E	Length of the ET lever arm at the ankle joint	0.06 m
ζ_{ET}	Angle of the ET lever arm at the ankle	80 deg
φ_0	Zero-position of the MACCEPA lever arm with respect to the shank	143 deg

The angle $\nu(t)$ can be written as:

$$\nu(t) = \arctan\left(\frac{y_F(t) - y_H(t)}{x_F(t) - x_H(t)}\right) \tag{A.13}$$

A.2.2. Energy transfer mechanism

The angle $\delta(t)$ can be written as:

$$\tan\delta(t) = \frac{E\cos\theta_{ET}(t) - p_{2x}}{p_{2y} + E\sin\theta_{ET}(t)} \tag{A.14}$$

A.3. Ankle–knee energy transfer module

The relationships between $d_1(t)$, $d_2(t)$, d_3 and the joint angles can be written as:

$$d_1(t) = 2\sqrt{\left(L_{sh} - p_{1y} - L_2(t)\right)^2 + \left(X_c - p_{1x} - L_2(t)\tan(\gamma_c)\right)^2} \tag{A.15}$$

$$d_2(t) = \sqrt{\left(p_{2y} + E\sin(\theta_{ET}(t))\right)^2 + \left(p_{2x} - E\cos(\theta_{ET}(t))\right)^2} \tag{A.16}$$

$$d_3 = \sqrt{\left(p_{1x} + p_{2x}\right)^2 + \left(p_{2y} - p_{1y}\right)^2} \tag{A.17}$$

where $L_2(t)$ depends on the knee angle $\theta_k(t)$ and $\theta_{ET}(t)$ depends on the ankle angle $\theta_a(t)$.

Appendix B. Supplementary table

The descriptions of the constant parameters used in the analytical model and their values are given in Table B.1.

Appendix C. Supplementary figures

The extended version of the geometrical model of the CYBERLEGs Beta-prosthesis in provided in Fig. C.1.

Fig. C.2 shows the piecewise polynomial interpolation of the MACCEPA angle $\alpha_M(t)$ as a function of the MACCEPA torque $\tau_{MAC}(t)$. The values of the polynomial coefficients a_P, b_P, c_P and d_P are written in the figure.

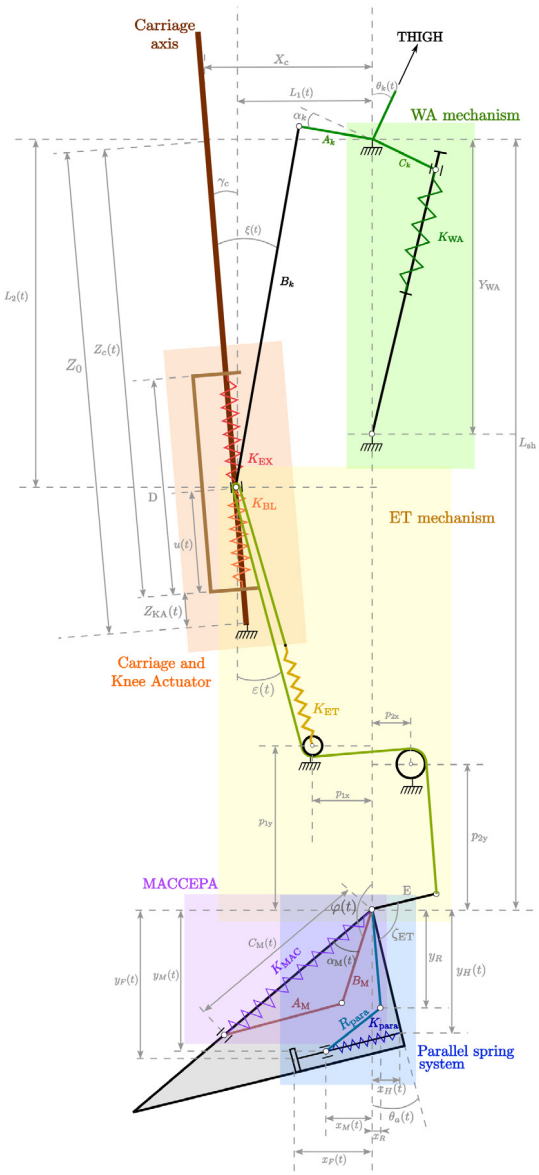


Fig. C.1. Extended version of the geometrical model of the CYBERLEGs Beta-prosthesis.

References

[1] Limb loss task force/amputee coalition (Knoxville, Tennessee), Roadmap for preventing limb loss in America: Recommendations from the 2012 limb loss task force, 2012.

[2] K. Ziegler-Graham, E.J. MacKenzie, P.L. Ephraim, T. Travison, R. Brookmeyer, Estimating the prevalence of limb loss in the United States: 2005 to 2050, Arch. Phys. Med. Rehabil. 89 (2008) 422–429.

[3] American Physical Therapy Association (APTA), Physical therapist's guide to above-knee amputation, Website https://www.moveforwardpt.com/SymptomsConditionsDetail.aspx?cid=7e9549ef-0bff-4b50-88f1-8a8bf4f1e496/ (Accessed April 2018).

[4] ESC guidelines on the diagnosis and treatment of peripheral artery diseases, Eur. Heart J. 32 (22) (2011) 2851–2906.

[5] European guidelines on cardiovascular disease prevention in clinical practice, Atherosclerosis 223 (1) (2012) 1–68.

[6] R. Waters, J. Perry, D. Antonelli, H. Hislop, Energy cost of walking of amputees: The influence of level of amputation, J. Bone Joint Surg. Amer. 58 (1976) 42.

[7] R.L. Waters, S. Mulroy, The energy expenditure of normal and pathologic gait, Gait Posture 9 (1999) 207–231.

[8] E.J. Rouse, R.D. Gregg, L.J. Hargrove, J.W. Sensinger, The difference between stiffness and quasi-stiffness in the context of biomechanical modeling, IEEE Trans. Biomed. Eng. 60 (2013) 562–568.

[9] A.H. Hansen, D.S. Childress, S.C. Miff, S.A. Gard, K.P. Mesplay, The human ankle during walking: implications for design of biomimetic ankle prostheses, J. Biomech. 37 (2004) 1467–1474.

[10] D.A. Winter, The Biomechanics and Motor Control of Human Gait: Normal, Elderly and Pathological, Waterloo Biomechanics Press, 1991.

[11] P. Cherelle, V. Grosu, L. Flynn, K. Junius, M. Moltedo, B. Vanderborght, D. Lefeber, The ankle mimicking prosthetic foot 3-locking mechanisms, actuator design, control and experiments with an amputee, Robot. Auton. Syst. 91 (2017) 327–336.

[12] M. Eilenberg, H. Geyer, H. Herr, Control of a powered ankle-foot prosthesis based on a neuromuscular model, IEEE Trans. Neural Syst. Rehabil. Eng. 18 (2010) 164–173.

[13] BionX, BionX's prosthesis or emPOWER, Website http://www.bionxmed.com/. (Accessed January 2018).

[14] R. Bellman, M. Holgate, T. Sugar, SPARKy 3: Design of an active robotic ankle prosthesis with two actuated degrees of freedom using regenerative kinetics, in: IEEE International Conference on Biomedical Robotics and Biomechatronics 2008, pp. 511–516.

[15] Össur, Rheo Knee, Website https://www.ossur.com/prosthetic-solutions/products/dynamic-solutions/rheo-knee.(Accessed January 2018).

[16] Ottobock, C-Leg above knee prosthetic leg, Website http://www.ottobockus.com/C-Leg.html (Accessed January 2018).

[17] Össur, Power Knee, Website https://www.ossur.com/prosthetic-solutions/products/dynamic-solutions/power-knee (Accessed January 2018).

[18] S. Pfeifer, A. Pagel, R. Riener, H. Vallery, Actuator with angle-dependent elasticity for biomimetic transfemoral prostheses, IEEE/ASME Trans. Mechatronics 20 (2015) 1384–1394.

[19] E.J. Rouse, L.M. Mooney, H.M. Herr, Clutchable series-elastic actuator: Implications for prosthetic knee design, Int. J. Robot. Res. 33 (2014) 1611–1625.

[20] Össur, Symbionic leg 3, Website http://www.ossur.com/prosthetic-solutions/products/dynamic-solutions/symbionic-leg (Accessed January 2018).

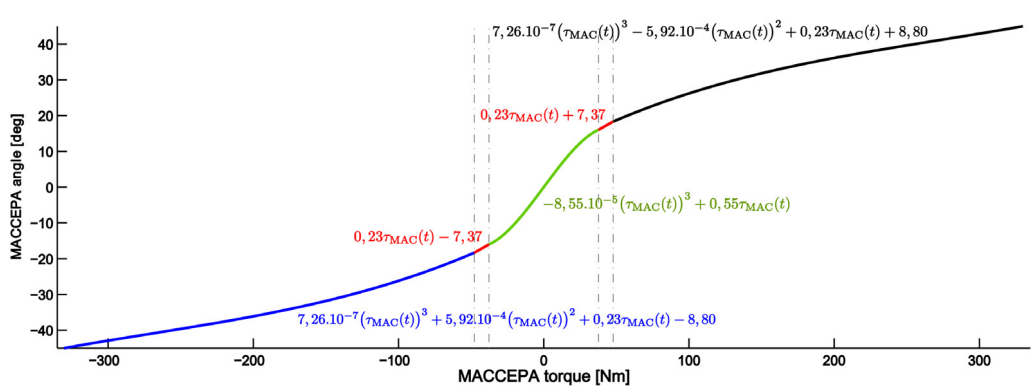


Fig. C.2. Piecewise polynomial interpolation of the MACCEPA angle $\alpha_M(t)$ as a function of the MACCEPA torque $\tau_{MAC}(t)$.

[21] Endolite, Linx, Website http://www.endolite.com/products/linx (Accessed January 2018).
[22] B.E. Lawson, J. Mitchell, D. Truex, A. Shultz, E. Ledoux, M. Goldfarb, A robotic leg prosthesis: Design control and implementation, IEEE Robot. Autom. Mag. 21 (2014) 70–81.
[23] F. Sup, H. Varol, M. Goldfarb, Upslope walking with a powered knee and ankle prosthesis: initial results with an amputee subject, IEEE Trans. Neural Syst. Rehabil. Eng. 19 (2011) 71–79.
[24] A. Matthys, P. Cherelle, M.V. Damme, B. Vanderborght, D. Lefeber, Concept and design of the HEKTA (Harvest Energy from the Knee and Transfer it to the Ankle) transfemoral prosthesis, in: Biomedical Robotics and Biomechatronics, BioRob, 2012 4th IEEE RAS EMBS International Conference on 2012.
[25] R. Unal, F. Klijnstra, B. Burkink, S. Behrens, E. Hekman, S. Stramigioli, H. Koopman, R. Carloni, Modeling of WalkMECH: A fully-passive energy-efficient transfemoral prosthesis prototype, in: Rehabilitation Robotics, ICORR, 2013 IEEE International Conference on 2013.
[26] L. Flynn, J. Geeroms, R. Jimenez-Fabian, B. Vanderborght, N. Vitiello, D. Lefeber, Ankle-knee prosthesis with active ankle and energy transfer: Development of the CYBERLEGs Alpha-prosthesis, Robot. Auton. Syst. 73 (2015) 4–15.
[27] J. Geeroms, L. Flynn, R. Jimenez-Fabian, B. Vanderborght, D. Lefeber, Energetic analysis and optimization of a maccepa actuator in an ankle prosthesis: energetic evaluation of the cyberlegs alpha-prosthesis variable stiffness actuator during a realistic load cycle, Auton. Robots 42 (2018) 147–158.
[28] J. Geeroms, L. Flynn, R. Jimenez-Fabian, B. Vanderborght, D. Lefeber, Design and energetic evaluation of a prosthetic knee joint actuator with a lockable parallel spring, Bioinspiration Biomimetics 12 (2017).
[29] M.R. Tucker, J. Olivier, A. Pagel, H. Bleuler, M. Bouri, O. Lambercy, J. Millán, R. Riener, H. Vallery, R. Gassert, Control strategies for active lower extremity prosthetics and orthotics: a review, J. NeuroEng. Rehabil. 12 (2015).
[30] D. Grimes, An Active Multimode Above Knee Prosthesis Controller (Ph.D thesis), Massachusetts Institute of Technology, Dept. of Mechanical Engineering, 1979 http://hdl.handle.net/17211/15998.
[31] H. Vallery, R. Ekkelenkamp, H. van der Kooij, M. Buss, Complementary limb motion estimation based on interjoint coordination: experimental evaluation, in: Proc. IEEE International Conference on Rehabilitation Robotics, ICORR, 2007, pp. 798–803. http://dx.doi.org/10.1109/ICORR.2007.4428516.
[32] B. Lawson, H. Varol, M. Goldfarb, Standing stability enhancement with an intelligent powered transfemoral prosthesis, IEEE Trans. Biomed. Eng. 58 (2011) 2617–2624.
[33] L. Ambrozic, M. Gorsic, J. Geeroms, L. Flynn, R.M. Lova, R. Kamnik, M. Munih, N. Vitiello, CYBERLEGs: A user-oriented robotic transfemoral prosthesis with whole-body awareness control, IEEE Robot. Autom. Mag. 21 (2014) 82–93.
[34] F. Sup, A. Bohara, M. Goldfarb, Design and control of a powered transfemoral prosthesis, Int. J. Robot. Res. 27 (2008) 263–273.
[35] F. Sup, H.A. Varol, J. Mitchell, T.J. Withrow, M. Goldfarb, Self-contained powered knee and ankle prosthesis: Initial evaluation on a transfemoral amputee, in: 2009 IEEE International Conference on Rehabilitation Robotics, 2009, pp. 638–644.
[36] V.R. Garate, A. Parri, T. Yan, M. Munih, R.M. Lova, N. Vitiello, R. Ronsse, Walking assistance using artificial primitives: a novel bioinspired framework using motor primitives for locomotion assistance through a wearable cooperative exoskeleton, IEEE Robot. Autom. Mag. 23 (2016) 83–95.
[37] V.R. Garate, A. Parri, T. Yan, M. Munih, R.M. Lova, N. Vitiello, R. Ronsse, Experimental validation of motor primitive-based control for leg exoskeletons during continuous multi-locomotion tasks, Front. Neurorobot. 11 (2017) 15.
[38] R. Jimenez-Fabian, L. Flynn, J. Geeroms, N. Vitiello, B. Vanderborght, D. Lefeber, Sliding-bar MACCEPA for a powered ankle prosthesis, J. Mech. Robot. 7 (2015) 1–2.
[39] C. Everarts, B. Dehez, R. Ronsse, Variable stiffness actuator applied to an active ankle prosthesis: Principle, energy-efficiency, and control, in: Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, 2012, pp. 323–328. http://dx.doi.org/10.1109/IROS.2012.6385789.
[40] B. Vanderborght, R.V. Ham, D. Lefeber, T. Sugar, K. Hollander, Comparison of mechanical design and energy consumption of adaptable, passive-compliant actuators, Int. J. Robot. Res. 28 (2009) 90–103.
[41] B. Vanderborght, N. Tsagarakis, R.V. Ham, I. Thorson, D. Caldwell, MACCEPA 2.0: Compliant actuator used for energy efficient hopping robot chobino1d, Auton. Robots 31 (2011) 55–65.
[42] A.D. Segal, M.S. Orendurff, G.K. Klute, M.L. McDowell, J.A. Pecoraro, J. Shofer, J.M. Czerniecki, Kinematic and kinetic comparisons of transfemoral amputee gait using C-Leg and Mauch SNS prosthetic knees, J. Rehabil. Res. Dev. 43 (2006) 857–870.
[43] M.A. Holgate, T.G. Sugar, A.W. Bohler, A novel control algorithm for wearable robotics using phase plane invariants, in: 2009 IEEE International Conference on Robotics and Automation, 2009, pp. 3845–3850. http://dx.doi.org/10.1109/ROBOT.2009.5152565.
[44] H. Lee, E.J. Rouse, H.I. Krebs, Summary of human ankle mechanical impedance during walking, IEEE J. Transl. Eng. Health Med. 4 (2016) 1–7.
[45] M.R. Tucker, A. Moser, O. Lambercy, J. Sulzer, R. Gassert, Design of a wearable perturbator for human knee impedance estimation during gait, in: 2013 IEEE 13th International Conference on Rehabilitation Robotics, ICORR, 2013, pp. 1–6. http://dx.doi.org/10.1109/ICORR.2013.6650372.
[46] S. Pfeifer, R. Riener, H. Vallery, Knee stiffness estimation in physiological gait, in: 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2014, pp. 1607–1610. http://dx.doi.org/10.1109/EMBC.2014.6943912.
[47] H. Huang, D.L. Crouch, M. Liu, G.S. Sawicki, D. Wang, A cyber expert system for auto-tuning powered prosthesis impedance control parameters, Ann. Biomed. Eng. 44 (2016) 1613–1624.

Sophie Heins received her M.S. in Biomedical Engineering from Université catholique de Louvain (Belgium) in 2013. She has been working as a Ph.D. student in the Center for Research in Mechatronics at UCL since September 2014.

Louis Flynn received his M.S. in Mechanical Engineering from Michigan State University in 2009. He is currently a Ph.D. student at the Vrije Universiteit Brussel working on actuated prostheses and exoskeletons design and control.

Joost Geeroms received his M.S. and Ph.D. in Engineering from Vrije Universiteit Brussel (Belgium), in 2011 and 2017. He is currently working on the CYBERLEGs++ project as a Post-doc researcher. His research is centred on actuated prosthetics and exoskeletons.

Dirk Lefeber received the degree in Civil Engineering at the Vrije Universiteit Brussel, and in 1986, earned a Ph.D. in Applied Sciences, Vrije Universiteit Brussel. Currently, he is Professor at the Dept. of Mechanical Engineering and head of the Robotics and Multibody Mechanics Research Group. His research interests are new actuators with adaptable compliance, dynamically balanced robots, robot assistants, rehabilitation robotics and multibody dynamics.

Renaud Ronsse obtained the M.S. (2002) and Ph.D. (2007) degrees in engineering from the Université de Liége. After post-docs at KULeuven and EPFL, he obtained an Assistant Professor position at UCLouvain in 2011, within the Institute of Mechanics, Materials, and Civil Engineering. Renaud Ronsse is interested in the transfer of principles governing neural control of human movements into human-centred robotics. His main application fields are bio-inspired assistive and rehabilitation robotics, mainly for the lower-limb, and humanoid robotics. More recently, he started a collaborative research to unveil the efficiency optimization mechanisms deployed by biological swimmers and flyers.